Tavily Research Agent

Pre-meeting briefings from the last 30 days, with sources you can check

Geek Room Global AI Hackathon

Theme: Tavily Research Agent (Tavily-primary)

November 2026

A real, runnable Python app: CLI plus web UI, Markdown and PDF output

1. Problem: meetings start cold

Before an important meeting, you need the last 30 days, fast

What did the company do lately: news, launches, hires, funding?

Manual search is slow, and it is easy to miss the one update that matters.

Plain AI answers mix old and new information, with no sources to check.

So people walk into sales, hiring, partnership, and investor meetings unprepared.

2. Solution: one page, in seconds

Type a name and a meeting date, get a briefing you can trust

A short summary of the last 30 days.

Key developments, each with a date and a source link.

3 to 5 discussion points and 3 to 5 open questions.

No invented facts: every development comes from a Tavily result and carries its source URL.

No results? The briefing says so clearly and lists nothing.

3. Demo flow: what judges will see

Four searches, one briefing, every source clickable

1. Open the CLI, the standard library web UI, or the Streamlit web UI.
2. Enter a real company name and a meeting date.
3. Four Tavily searches run: news, launches, hires, funding.
4. Briefing appears: newest first, duplicates removed, old results dropped.
5. Open one source link to prove it is real.
6. Markdown and PDF exports, same content in both.

4. Technical architecture

A small pipeline with the 30-day rule enforced twice

Input: CLI, Python standard-library web UI, or Streamlit web UI (app.py), all one pipeline.

Tavily Search API: POST api.tavily.com/search, Bearer auth, topic news, time_range month, start_date and end_date for the 30-day window.

Parse published_date (ISO and RFC/GMT), filter outside the window, deduplicate by normalised URL, sort newest first, undated flagged.

Summarise: relevance-aware extractive by default, optional LLM. Render: Markdown and PDF.

Key is never hardcoded: Tavily key comes only from TAVILY_API_KEY.

Optional: Tavily Extract adds full content for the top 8 developments, off by default. Streamlit demo mode shows the prebuilt live briefings with no key.

5. Live proof: real Tavily data, Nov 5, 2026

12 searches, 60 results, 37 kept, every kept item sourced

Meeting date 2026-11-05, window 2026-10-06 to 2026-11-05.

OpenAI: 14 kept of 20. Google: 15 kept of 20. Microsoft: 8 kept of 20.

Dropped 23: 16 truncated URL, 5 profile pages, 2 duplicates, 0 too old.

Files: examples/live-openai/, examples/live-google/, examples/live-microsoft/, briefing.md and briefing.pdf in each, plus the source JSON.

Tests: 74 run, 74 passed (offline unit tests, all HTTP mocked).

Briefing quality measured in QUALITY.md: 37 of 37 dated, 37 of 37 sourced, 18 of 37 name the subject in both title and snippet.

6. Live findings, from the kept results

Real headlines, real dates, real links in the briefings

OpenAI: to acquire Pierre Computer Company, Nov 4, 2026.

OpenAI: GPT-6 and Intelligent UI in ChatGPT, Oct 8, 2026.

Google: Gemini AI workplace agent that can write code and run tasks, Oct 8, 2026.

Google: universal online shopping cart ahead of the holidays, Oct 15, 2026.

Microsoft: buyouts to 60,000 workers as it pours money into AI, Oct 26, 2026.

Microsoft: plans to raise $25 billion in debt for the AI buildout, Oct 30, 2026.

7. How AI is used

Tavily does the research, the summary cannot invent facts

Tavily Search is the research layer: real-time results, ranked, with snippets.

Summarisation is extractive by default: the summary reuses the first sentences of the top result snippets and counts results by category.

The summary is relevance-aware: only developments that name the subject in both title and snippet are quoted in the Top findings. Other kept developments stay listed in full, with source links.

An LLM summary is used only if LLM_API_KEY is set, only on supplied snippets, told not to invent dates, facts, or sources, with fallback to extractive.

The briefing labels which mode was used (extractive or llm).

The LLM is never used to add new developments.

8. Impact and scalability

One briefing per meeting, for any company or person

Saves preparation time before every sales, hiring, partnership, or investor meeting.

Works for any company or person with public web coverage.

Scales by adding categories, languages, or export formats, without changing the core filter and sourcing rules.

Cost is bounded: 4 searches per briefing, basic depth by default. Tavily Extract is off by default and enriches only the top 8, about 8 credits for 37 URLs.

9. Limitations, next steps, and why this wins

Finished, working, and honest

Playground snippets are truncated at about 100 characters, kept exactly as returned.

The direct API route is still unproven: HTTP 401 here, while the same key works in the Playground. The live briefings prove the pipeline on real Tavily data.

Tavily Extract is also unproven live, and the Streamlit app is not deployed yet. Both are code and mocked tests, off by default.

Submitted for evaluation Nov 5, 2026, Under Review. The repo stays open for improvements until judging on Nov 29, 2026.

Judges can run it and check every source in the live briefings.

The 30-day rule is enforced in code and in tests, not just in words.

Tavily Research Agent: know what happened, before you walk in.